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CARBON DIOXIDE GAS INSULATED SWITCHGEAR

Abstract

A switchgear apparatus may include a cylindrical housing defining a housing interior filled with carbon dioxide, an electrical switch supported within the housing, the electrical switch including a chamber communicating the housing interior and a pair of contacts within the chamber, the pair of contacts including a stationary contact and a movable contact, and a crankshaft coupled to the movable contact such that the movable contact is configured to translate linearly relative to the stationary contact in response to rotation of the crankshaft.

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Background/Summary

FIELD

[0001] The present disclosure relates to electrical switchgear.

BACKGROUND

[0002] Electrical switchgear is used to selectively open and close electrical connections in electrical power generation, transmission, and distribution systems. Gas-insulated electrical switchgear typically uses sulfur hexafluoride (SF₆) contained within a tank to surround and insulate the high voltage components of the switchgear.

SUMMARY

[0003] SF₆ is known to be a greenhouse gas that contributes to global warming, and its use as an arc-quenching insulating gas in switchgear is being phased out in some markets and regions. Accordingly, a need exists for switchgear insulated with alternative (i.e., non-SF₆) gases, and particularly natural origin gases, such as carbon dioxide ("CO₂"), Nitrogen, atmospheric air, or the like. CO₂ or other non-SF₆ gases can be used as a replacement for SF₆ as the insulating gas, however, such non-SF₆ gases may need to be contained in the switchgear housing at a higher pressure than if SF₆ were used. In addition, non-SF₆ gases may be less effective at arc-quenching than SF₆ in rotary switches. CO₂ has been found effective for arc-quenching in linear switchgear, but linear switchgear often has a larger footprint than rotary switchgear. Due to its larger footprint, typical linear switchgear may not fit into an existing location (e.g., cabinet, underground vault, etc.) for a user who wishes to replace rotary switchgear insulated using SF₆ with switchgear insulated using CO₂ or other non-SF₆ gases.

[0004] Accordingly, the present disclosure provides, among other things, a switchgear apparatus insulated with a non-SF₆ gas, such as CO₂, and enclosed within a cylindrical housing. The switchgear apparatus includes linearly movable contacts actuated via a rotary mechanism. This may advantageously allow the switchgear apparatus to fit into an existing rotary switchgear installation.

[0005] For example, in some aspects, the techniques described herein relate to a switchgear apparatus including: a cylindrical housing defining a housing interior that contains carbon dioxide; an electrical switch supported within the housing, the electrical switch including a chamber communicating the housing interior and a pair of contacts within the chamber, the pair of contacts including a stationary contact and a movable contact; a crankshaft coupled to the movable contact such that the movable contact is configured to translate linearly relative to the stationary contact in response to rotation of the crankshaft.

[0006] In some aspects, the techniques described herein relate to a switchgear apparatus, wherein the electrical switch is a first electrical switch, and wherein the switchgear apparatus further includes a second electrical switch supported within the housing.

[0007] In some aspects, the techniques described herein relate to a switchgear apparatus, wherein the crankshaft is a first crankshaft, and wherein the switchgear apparatus further includes a second crankshaft coupled to a movable contact of the second electrical switch.

[0008] In some aspects, the techniques described herein relate to a switchgear apparatus, wherein the first crankshaft is rotatable in a first direction to move the movable contact of the first electrical switch away from the stationary contact of the first electrical switch, and wherein the second crankshaft is rotatable in a second direction opposite the first direction to move the movable contact of the second electrical switch away from a stationary contact of the second electrical switch.

[0009] In some aspects, the techniques described herein relate to a switchgear apparatus, wherein the housing is filled with carbon dioxide to a gauge pressure of at least 15 psi.

[0010] In some aspects, the techniques described herein relate to a switchgear apparatus, wherein the housing is filled with carbon dioxide to a gauge pressure of at least 20 psi.

[0011] In some aspects, the techniques described herein relate to a switchgear apparatus including: an apparatus housing including a base casting coupled to a shell, a top plate, and a base plate, the top plate and base plate coupled to opposite ends of the base casting; a crankshaft rotationally supported between the top plate and the base plate; an operation assembly coupled to the base plate and configured to provide a rotational output to the crankshaft; a plurality of chambers coupled to the base casting and extending from the base casting away from the shell, each of the chambers containing an insulating medium; a top support coupled to distal ends of the chambers, the top support configured to seal the distal ends of the chambers; a plurality of contact pairs supported in each chamber, each contact pair including a stationary contact and a movable contact, each stationary contact is coupled to the top support and each movable contact is coupled to the crankshaft, each movable contact configured to translate linearly along the chamber in response to rotation of the crankshaft.

[0012] In some aspects, the techniques described herein relate to a switchgear apparatus, wherein the operation assembly includes an input shaft coupled to the crankshaft and configured to provide the rotational output to the crankshaft, the input shaft configured to receive a rotational input from an operator, the rotational input having a first direction.

[0013] In some aspects, the techniques described herein relate to a switchgear apparatus, wherein the operation assembly further includes a transmission assembly configured to receive the rotational input in the first direction and provide the rotational output to the crankshaft in a second direction opposite to the first direction.

[0014] In some aspects, the techniques described herein relate to a switchgear apparatus, wherein the transmission assembly includes a pinion coupled to the input shaft and a driven gear coupled to an output shaft, the output shaft coupled to the crankshaft.

[0015] In some aspects, the techniques described herein relate to a switchgear apparatus including: a housing defining a housing interior containing an insulating medium; a lid coupled to the housing and the lid defining an interior; a plurality of connectors extending from the lid into the interior; and an electrical switch coupled to the lid and electrically coupled to the connectors, the electrical switch including an apparatus housing including a base casting coupled to a shell, a top plate, and a base plate, the top plate and base plate coupled to opposite ends of the base casting, a crankshaft rotationally supported between the top plate and the base plate, an operation assembly coupled to the base plate and configured to provide a rotational output to the crankshaft, a plurality of chambers coupled to the base casting and extending from the base casting away from the shell, each of the chambers communicating the housing interior, a top support coupled to distal ends of the chambers, the top support configured to seal the distal ends of the chambers, and a plurality of contact pairs supported in each chamber, each contact pair including a stationary contact and a movable contact, each stationary contact is coupled to the top support and each movable contact is coupled to the crankshaft, each movable contact configured to translate linearly along the chamber in response to rotation of the crankshaft.

[0016] In some aspects, the techniques described herein relate to a switchgear apparatus, wherein the operation assembly includes a transmission assembly configured to receive a rotational input in a first direction and provide the rotational output to the crankshaft in a second direction opposite to the first direction.

[0017] In some aspects, the techniques described herein relate to a switchgear apparatus, wherein the transmission assembly includes a pinion coupled to an input shaft and a driven gear coupled to an output shaft, the output shaft coupled to the crankshaft.

[0018] Other features and aspects of the disclosure will become apparent by consideration of the following detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 is a perspective view of an embodiment of an electrical switchgear apparatus.

[0020] FIG. 2 is a side view of an actuator of the electrical switchgear apparatus of claim 1.

[0021] FIG. 3 is a perspective view of an electrical switchgear apparatus of FIG. 1, including a lockout assembly.

[0022] FIG. 4 is a perspective view of another embodiment of an electrical switchgear apparatus, including a lockout assembly.

[0023] FIG. 5 is a perspective view of the electrical switchgear apparatus of FIG. 1, including the electrical switch.

[0024] FIG. 6 is an exploded view of an embodiment of an electrical switch.

[0025] FIG. 7 is a perspective view of the electrical switch of FIG. 6.

[0026] FIG. 8 is a side view of the electrical switch of FIG. 6.

[0027] FIG. 9 is a section view of the electrical switch of FIG. 6.

[0028] FIG. 10 is a perspective view of an embodiment of an operation mechanism.

[0029] FIG. 11 is a perspective view of an embodiment of the spring assembly.

[0030] FIG. 12 is a perspective view of the spring assembly of FIG. 11.

[0031] FIG. 13 is a section view of the spring assembly of FIG. 11.

[0032] FIG. 14 is a perspective view of another embodiment of an operation mechanism, including a transmission assembly.

[0033] FIG. 15 is a perspective view of the operation mechanism of FIG. 14.

[0034] FIG. 16 is a perspective view of a portion of the electrical switch of FIG. 6.

[0035] FIG. 17 is a perspective view of an embodiment of a crankshaft.

[0036] FIG. 18 is a circuit diagram of an embodiment of a switchgear apparatus.

[0037] FIG. 19 is an end view of a portion of an embodiment of a switchgear apparatus including the circuit of FIG. 18.

[0038] FIG. 20 is a perspective view of a portion of the switchgear apparatus of FIG. 19.

[0039] FIG. 21 is a circuit diagram of an embodiment of a switchgear apparatus.

[0040] FIG. 22 is an end view of a portion of an embodiment of a switchgear apparatus including the circuit of FIG. 21.

[0041] FIG. 23 is a perspective view of a portion of the switchgear apparatus of FIG. 22.

[0042] FIG. 24 is a circuit diagram of an embodiment of a switchgear apparatus.

[0043] FIG. 25 is an end view of a portion of an embodiment of a switchgear apparatus including the circuit of FIG. 24.

[0044] FIG. 26 is a perspective view of a portion of the switchgear apparatus of FIG. 25.

[0045] FIG. 27 is a circuit diagram of an embodiment of a switchgear apparatus.

[0046] FIG. 28 is an end view of a portion of an embodiment of a switchgear apparatus including the circuit of FIG. 27.

[0047] FIG. 29 is a perspective view of a portion of the switchgear apparatus of FIG. 28.

[0048] FIG. 30 is a circuit diagram of an embodiment of a switchgear apparatus.

[0049] FIG. 31 is an end view of a portion of an embodiment of a switchgear apparatus including the circuit of FIG. 30.

[0050] FIG. 32 is a perspective view of a portion of the switchgear apparatus of FIG. 31.

[0051] FIG. 33 is a circuit diagram of an embodiment of a switchgear apparatus.

[0052] FIG. 34 is an end view of a portion of an embodiment of a switchgear apparatus including

the circuit of FIG. 33.

[0053] FIG. 35 is a perspective view of a portion of the switchgear apparatus of FIG. 34.

[0054] FIG. 36 is a circuit diagram of an embodiment of a switchgear apparatus.

[0055] FIG. 37 is an end view of a portion of an embodiment of a switchgear apparatus including the circuit of FIG. 36.

[0056] FIG. 38 is a perspective view of a portion of the switchgear apparatus of FIG. 37.

[0057] Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

[0058] Terms of approximation, such as “generally,” “approximately,” or “substantially,” include values within ten percent greater or less than the stated value. When used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction. For example, “generally vertical” includes directions within ten degrees of vertical in any direction, e.g., clockwise or counter-clockwise.

[0059] Benefits, other advantages, and solutions to problems are described below with regard to specific embodiments. However, the benefits, advantages, solutions to problems, and any feature(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential feature of any or all the claims.

DETAILED DESCRIPTION

[0060] FIG. 1 illustrates an embodiment of an electrical switchgear apparatus 10 that includes a lid 14 coupled to the first end 18 of a substantially cylindrical switchgear housing 22. The electrical switchgear apparatus 10 also includes a foundation 26 coupled to a second end 30 of the housing 22 opposite the first end 18. The illustrated foundation 26 includes supports (e.g., rails) 34 coupled to a circular base plate 38; however, the foundation 26 may have other configurations to suit a particular installation location. The lid 14, housing 22, and circular base plate 38 together define a housing interior I (FIG. 5) in which an insulating medium (e.g., CO₂ gas, a combination of CO₂ and oxygen (“O₂”), and/or other non-SF₆ gases) is contained. The housing interior I has a volume of less than 500 liters, and may have a volume that is less than 300 liters, for instance, a volume that is 150 liters, or less.

[0061] With continued reference to FIG. 1, the illustrated switchgear apparatus 10 includes line couplers 42 on the lid 14, to which supply and load lines of the supply and load circuits may be coupled. In some embodiments, the switchgear apparatus 10 may include two, four, six, or more line couplers depending on the number of phases and ways present in the circuit in which the electrical switchgear apparatus 10 is used. The embodiment of FIG. 1 and other embodiments described below may include input couplers 46 that may receive or otherwise connect to a fuse, a voltage tap, or other control or diagnostic structure therein.

[0062] As described in greater detail below, embodiments of the switchgear apparatus 10 described and illustrated herein may include, among other things, an electrical switch having a fixed contact electrically connected to one of a line coupler or a load coupler, and a movable contact electrically connected to the other of the line coupler and the load coupler. An actuator provides a rotational input to an operation assembly, which converts rotational movement of the actuator into linear movement of the movable contact to open or close the circuit between the line coupler and the load coupler. The electrical switchgear apparatus 10 is therefore a rotary-actuated linear switchgear apparatus. The operation assembly may include an energy storage assembly to store energy (e.g., in springs) as the actuator rotates, and then abruptly release the stored energy to rapidly drive the movable contact. A lockout assembly may be provided to prevent access to control structures, such as fuses, within the line coupler or load coupler when the contacts are closed.

[0063] In more detail, the illustrated switchgear apparatus **10** includes an actuator **50** rotationally supported by the lid **14** and configured to control the electrical output of the switchgear apparatus **10** to the load circuits by opening or closing one or more electrical switches **200** (embodiments of which are shown starting at FIG. 5 and described in further detail below). Illustrated in FIG. 2, the actuator **50** includes an actuator shaft **54** that extends through a boss **58** fixed (e.g., welded) to the lid **14**. The illustrated actuator **50** also includes a seal in the form of O-rings **62** (e.g., two O-rings) disposed on the actuator shaft **54** between the actuator shaft **54** and the boss **58** to maintain the interior of the electrical switchgear apparatus **10** in a sealed state relative to the exterior of the electrical switchgear apparatus **10**. Snap rings **66** and washers **70** maintain the position of the actuator shaft **54** within the boss **58**. An indicator **74** is positioned on the actuator shaft **54** toward the external end **78** of the actuator shaft **54** and a nut **82** is coupled to the external end **78**. The indicator **74** and nut **82** are coupled with the actuator shaft **54** for rotation with the actuator shaft **54**. In the present embodiment, the indicator **74** and actuator shaft **54** are coupled by a matching D-profile, although other coupling methods may be used instead (e.g., by a keyway, welding, etc.). The illustrated actuator **50** also includes a switch coupling **86** coupled to the internal end **90** of the actuator shaft **54** and to the electrical switch **200** (e.g., at the input shaft **216**, also shown in FIG. 5) by a pair of coupling pins **94** that extend outwardly from the actuator shaft **54** and input shaft **216** through slots **98** in the switch coupling **86**. To operate, that is, open or close, the electrical switch **200** to which the actuator **50** is coupled, a tool (e.g., a wrench or T-handle with a socket matching the profile of the nut **82**) is placed on the nut **82** and rotated. In some embodiments, the actuator **50** may be provided with a handle.

[0064] With reference to FIGS. 3-4, in some embodiments, the electrical switchgear apparatus **10** may include a lockout assembly **134** coupled to the lid **14**. In the embodiment of FIG. 3 (e.g., a 1-phase, fused electrical switchgear apparatus **10**), the lockout assembly **134** includes a lockout plate **138** that is pivotably coupled to the lid **14** and selectively disposed above the input couplers **46** to at least partially cover the input couplers **46**, thereby preventing access to remove the control structure (e.g., fuse) within each input coupler **46**. A lockout bar **142** is coupled to the nut **82** on the actuator shaft **54** for rotation with the actuator shaft **54**. In the present embodiment, the lockout bar **142** is integrally formed with the indicator **74** as a unitary piece coupled to the actuator shaft **54**. In other embodiments, the indicator **74** and lockout bar **142** may be separately formed.

[0065] In the embodiment of FIG. 4 (e.g., a 3-phase, fused electrical switchgear apparatus **10**), the electrical switchgear apparatus **10** includes two electrical switches **200**, and each electrical switch **200** includes an actuator **50** to which a lockout assembly **134'** is coupled. Each a lockout assembly **134'** includes a lockout plate **138'** and a lockout bar **142'**. The lockout plate **138'** is pivotably coupled to the lid **14** and selectively disposed above the input couplers **46** to at least partially cover the input couplers **46**. Lockout plate extensions **146** extend from the lockout bar **142'**. The lockout bar **142'** of each lockout assembly **134'** is coupled to the actuator shaft **54** of each actuator **50** for rotation with the actuator shaft **54**.

[0066] Returning to FIGS. 3-4, when the actuator shaft **54** and the associated electrical switch **200** are in a closed position (e.g., current can flow through the electrical switch **200**), at least a portion of the lockout bar **142**, **142'** is positioned above the lockout plate **138**, **138'** and prevents rotation of the lockout plate **138**, **138'** to uncover the input couplers **46**. In the second embodiment of the lockout shaft of FIG. 4, the lockout bars **142'** are positioned above the lockout plate extensions **146**. When the actuator shaft **54**, and electrical switch **200** to which it is coupled, are in the open position (e.g., current cannot flow through the electrical switch **200**), the lockout bar **142**, **142'** does not cover the lockout plate **138**, **138'**, and the lockout plate **138**, **138'** can be pivoted to uncover the input couplers **46** and allow access for removal and/or replacement of the control structures (e.g., fuses) within the input couplers **46**. Other embodiments of lockout assemblies that physically prevent access to the input couplers **46** when the electrical switch **200** are in a closed position in which current can flow through the electrical switch **200** may be used instead. Other lockout

mechanisms may be used in addition to or instead of the lockout assemblies described. In the embodiments of FIGS. 3-4, the lockout bar **142**, **142'** includes a hole **143** that is alignable with the hole **144** of a safety plate **145**, **145'** coupled (e.g., welded, fastened, etc.) to the lid **14**. When the electrical switch **200** is in the open position, holes **143**, **144** are aligned and a lock (e.g., a padlock) can be inserted into the holes **143**, **144** and locked, preventing rotation of the lockout bar **142**, **142'**, that is, preventing inadvertent closing of the electrical switch **200**, for instance, while the electrical switch **200** is being serviced.

[0067] FIG. 5 illustrates an embodiment of the electrical switch **200** coupled to an interior side **150** of the lid **14** (illustrated schematically) and positioned within the interior **154** of the housing **22**. The line couplers **42** and input couplers **46** extend into the interior **154** and are electrically coupled to the electrical switch **200** by suitable connectors, such as copper wire rope, busbars, or the like. The illustrated electrical switch **200** includes an operation assembly **208** including a mounting plate **212** that is coupled to the lid **14** and an input shaft **216**. The operation assembly **208** is operationally engaged with the actuator shaft **54** of the actuator **50** by the switch coupling **86**, which is coupled to the input shaft **216**. The electrical switchgear apparatus **10** may include bushings **220** that extend inward into the interior **154** of the housing **22** and are electrically coupled to the line couplers **42** and to the electrical switch **200** (e.g., by a connector).

[0068] With reference to FIGS. 6-9, the operation assembly **208** of the illustrated electrical switch **200** is coupled to a base plate **224**. A shell **228** and a base casting **232** are coupled to and extend from the base plate **224**, and a top plate **236** is coupled to the shell **228** and base casting **232** opposite the base plate **224**. The base plate **224**, shell **228**, base casting **232**, and top plate **236** (coupled together by fasteners **238**, shown in FIG. 7) cooperate to define an interior **240** through which a crankshaft **244** extends. Plates **246** are positioned in the interior **240** of the shell **228** at intervals along the crankshaft **244**. The plates **246** each have a U-shaped cutout **247**. The crankshaft **244** extends and is supported between the base plate **224** and top plate **236** and is operationally coupled to the operation assembly **208** to rotate in response to an output from the operation assembly **208** (FIG. 5). One or more chambers **248** (e.g., three chambers, each corresponding with a phase of a three-phase circuit) are coupled to the base casting **232** on the opposite side of the base casting **232** from the shell **228** offset from the middle of the base casting **232**. The chambers **248** extend away from the base casting **232**. The plurality of chambers **248** are open to (i.e., communicate) the housing interior I and are therefore filled with the insulating medium (e.g., CO₂ gas, a combination of CO₂ and oxygen ("O₂"), and/or other gases) contained within the housing interior I. In some embodiments, the housing interior I and chambers **248** may contain the insulating medium at a gauge pressure of at least 15 psi, or at least 20 psi in some embodiments. In other embodiments, other insulating gases and/or other fill pressures may be used. A top support **252** is coupled to the distal ends of the plurality of chambers **248** to define an internal compartment **256** of each chamber **248**. The base plate **224**, shell **228**, base casting **232** may be composed of dielectric material that is not electrically conductive to prevent electrical discharge. Other shielding may be included to cover conductive components and prevent arcing or other electrical discharge.

[0069] The electrical switch **200** includes pairs of contacts **260** that correspond to the number of chambers **248** (e.g., three pairs of contacts) in the electrical switch **200**. Each pair of contacts **260** includes a first, stationary contact **264** that is coupled to the top support **252** and a second, movable contact **268** coupled to the crankshaft **244**. A puffer nozzle **272** is coupled to the movable contact **268**. Each stationary contact **264** is electrically coupled to one of the line couplers **42** or input couplers **46** by a connector **204** (shown in FIG. 5) to receive electrical current from a supply line or to transmit electrical current to a load line. With reference to FIG. 9, the movable contact **268** is linearly translatable within the chamber **248** between a closed position in which the movable contact **268** engages or is in contact with the stationary contact **264**, and an open position (puffer nozzle **272** illustrated schematically in FIG. 9) in which the movable contact **268** is spaced from the stationary contact **264**. When the movable contact **268** is in the closed position, current can flow

between the stationary contact **264** and the movable contact **268**.

[0070] As shown in FIGS. **7** and **9**, the electrical switch **200** also includes sliding contacts **276** that are coupled to (e.g., cast or molded in) the base casting **232** through which the movable contacts **268** extend and are slidable. The sliding contacts **276** are configured to engage the movable contact **268** for electrical current exchange between the movable contact **268** and the sliding contacts **276**. Each sliding contacts **276** includes a current exchange hub **278** and a tab portion **280** extending from the current exchange hub **278**. The tab portion **280** is positioned between the shell **228** and the base casting **232** to extend beyond the outer surface of the shell **228** and base casting **232** and be electrically coupled (e.g., by one of the connectors **204**) to one of the line couplers **42**, input couplers **46**, or another sliding contact **276** of a second electrical switch **200** in the electrical switchgear apparatus **10**. The sliding contacts **276** also include one or more sliding electrical contacts **284** (e.g., two sliding electrical contacts, or lamina) that engage the movable contact **268** and a sleeve bushing **288**. The plates **246** are positioned within the interior **240** of the shell **228**, with the crankshaft **244** disposed within the U-shaped cutout **247** of each plate **246**. The plates **246** electrically isolate the sliding contacts **276** from one another and from the base plate **224** and the top plate **236**. In the present embodiment, plates **246** are positioned adjacent the base plate **224** and top plate **236** and between each sliding contact **276**.

[0071] With reference to FIGS. **10-13**, the operation assembly **208** includes the mounting plate **212** and an energy storage assembly **290** (FIGS. **11-13**) coupled between the mounting plate **212** and the base plate **224**. A plurality of standoffs **294** are coupled between the mounting plate **212** and the base plate **224** (e.g., by fasteners). The mounting plate **212** includes a planar plate portion **302** from which a plurality of legs **306** (e.g., three legs) extends. Each of the legs **306** include a tab portion **310** that is coupled to the inside of the lid **14**. The plate portion **302** includes a hole **314** positioned substantially in the center of the plate portion **302** through which the input shaft **216** extends.

[0072] The energy storage assembly **290** receives a rotational input from the input shaft **216**, to compress springs **318**, **322** (e.g., as stored energy), which are supported in the energy storage assembly **290** by first and second plates **326**, **330** when the stationary contacts **264** and the movable contacts **268** are engaged to close the circuit. The input shaft **216** is supported in the first and second plates **326**, **330** by flanged bushings **334** and washers **338** (FIG. **13**). The energy storage assembly **290** includes pins **340** coupled to a center plate **342** positioned between the first and second plates **326**, **330** (FIG. **12**). The pins **340** extend through the base plate **224** and are coupled to the crankshaft **244**.

[0073] Rotation of the input shaft **216** in a first direction **346** about the rotational axis **344** to open the electrical switch **200** translates the movable contacts **268** from the closed position to the open position and releases the energy stored by the energy storage assembly **290** in the compressed springs **318**, **322**. Release of the energy stored in the springs rotates the pins **340** about the axis **344**, and thereby rotates the crankshaft **244** to translate the movable contacts **268** to the open position. There is no motion until there is sufficient energy to operate the crankshaft **244** at the required rotational speed. This is achieved using interlock assemblies **347** coupled to the pins **340** and the center plate **342**. The interlock assemblies **347** each include an interlock linkage **348** that engages one of the standoffs **294**. The standoffs **294** hold the interlock linkages **348** in place during motion (e.g., rotation) of the center plate **342**. As the center plate **342** pulls the interlock linkages **348** past the respective standoffs **294**, the energy stored in the springs **318**, **322** is released through the crankshaft **244**. In other words, the energy stored in the springs **318**, **322** when the stationary contacts **264** and movable contacts **268** are engaged to close the circuit assists to disengage the contacts **264**, **268** and quickly open the circuit. The same interlock assemblies **347** automatically re-engage on the standoffs **294** at the end of the operation. Rotation of the input shaft **216** about the axis **344** in a opposite direction about the rotational axis **344**, which corresponds with moving the movable contacts **268** into engagement with the stationary contacts **264** and closing the circuit, compresses the springs **318**, **322**.

[0074] FIGS. 14-16 illustrate another embodiment of an operation assembly 350 (e.g., a reverse operation assembly) that transfers rotation from the actuator 50 to the crankshaft 244 in a direction opposite to the rotational direction of the actuator 50. Components of the operation assembly 350 that are identical to those of the previous embodiment of the operation assembly 208 will not be described.

[0075] The operation assembly 350 includes a mounting plate 212 and an energy storage assembly 290 coupled between the mounting plate 212 and the base plate 224, and a plurality of standoffs 294 couple the mounting plate 212 to the base plate 224, substantially similar to operation assembly 208. The operation assembly 350 also includes a transmission assembly 352 (FIG. 15) positioned between the mounting plate 212 and a support plate 356 that is spaced from, and coupled, to the mounting plate 212 in a direction opposite the base plate 224 by standoffs 358. The support plate 356 is positioned between the plurality of legs 306 of the mounting plate 212. An input shaft 360 having coupling pins 94 is coupled to an actuator 50 and extends through the support plate 356. The transmission assembly 352 includes a drive pinion 364 that is coupled to the input shaft 360 for rotation with the input shaft 360 and a driven gear 368 supported on an output shaft 372 (e.g., a shaft similar to the input shaft 216). The output shaft 372 extends through the center of the support plate 356 and mounting plate 212 and is coupled to the energy storage assembly 290 to provide a rotational input to the energy storage assembly 290. The input shaft 360 is offset from the center of the support plate 356 and mounting plate 212.

[0076] The input shaft 360 receives a rotational input in a first direction 376 about the input shaft axis 380 from the actuator 50. The drive pinion 364 rotates with the input shaft 360, which in turn, rotates the driven gear 368 that is engaged (e.g., meshed) with the drive pinion 364. The driven gear 368 is thus rotated in a second direction 384 about the rotational axis 388 of the output shaft 372. Rotation of the output shaft 372 is transmitted to the energy storage assembly 290, which provides a rotational input to the crankshaft 244 in substantially the same manner as described above for the operation assembly 208. It will be appreciated that the operation assembly 350 can be used in place of operation assembly 208 to provide flexibility to optimize layout of the electrical switch 200 within the housing interior. It will also be appreciated that the operation assembly 350 can be included to communize operation electrical switchgear apparatus 10 having more than one electrical switch 200. For instance, in an embodiment of the electrical switchgear apparatus 10 having two electrical switches 200 (e.g., the embodiments of FIGS. 22, 25, 28, 31, described in greater detail below), by using one electrical switch 200 with operation assembly 208 and a second electrical switch 200 with operation assembly 350, when opening or closing the circuits, the operator can rotate lockout bars 142, 142' in the same rotational direction and the crankshafts 244 of each of the electrical switches 200 will rotate in opposite directions to open or close the circuits.

[0077] With reference to FIGS. 16-17, the crankshaft 244 is coupled to an operation assembly 208 (while not included in the embodiment, the crankshaft may instead be coupled to operation assembly 350). A first end 389 of the crankshaft 244 is positioned in a drive coupler 390 (e.g., fit within a bore 391 of the drive coupler 390 having a square cross-section). The drive coupler 390 includes flats 392 and a circumferential recess 393 formed on the outer circumference 394 of the drive coupler 390. The flats 392 selectively engage one or more switches (not shown) that provide a signal to indicate the position of the crankshaft 244 or other data. A plate 246 is partially positioned within the circumferential recess 393. The drive coupler 390 is coupled to the operation assembly 208, 350 by the pins 340 and driven by the operation assembly 208, 350 to rotate about the crankshaft axis 400. A bearing plate 395 is positioned on the second end 397 of the crankshaft 244. The bearing plate 395 includes a flange 398 that extends outward and engages the top plate 236 to position the bearing plate 395 and crankshaft 244 in the top plate 236. The drive coupler 390 and bearing plate 395 are composed of polypropylene or other dielectric material. Coupling plates 404 are spaced along the shaft portion 396 and are coupled to the movable contacts 268 by pins

408.

[0078] The electrical switch **200** is assembled by coupling the base plate **224** and operation assembly **208** with the base casting **232**, positioning the drive coupler **390** and bearing plate **395** on the crankshaft **244**, coupling the crankshaft **244** and drive coupler **390** to the base plate **224**, positioning the top plate **236** with the bearing plate **395** to fit the crankshaft **244** and bearing plate **395** between the top plate **236** and base plate **224**, coupling the top plate **236** to the base casting **232** (e.g., with four fasteners), positioning the plates **246** along the crankshaft **244**, and coupling the shell **228** to the base casting **232**. It will be appreciated that by assembling the components as described, the number of components to be assembled is reduced and the number of fastening operations required is also reduced, thereby reducing the manufacturing time.

[0079] FIGS. **18-38** illustrate embodiments of different electrical switchgear apparatuses.

Depending on the electrical layout of each embodiment, the switchgear assembly has either one or two electrical switch assemblies. The electrical switch assemblies are coupled to connectors extending inward into the interior of the housing by different busbars (e.g., a copper wire rope, a copper bus with insulation, a 2-layer bus, lamina-bus assemblies, etc.).

[0080] FIGS. **18-20** illustrate an embodiment of an electrical switchgear apparatus **1010** (e.g., a 1-phase, fused electrical switchgear apparatus), with like components and features as the embodiments of the electrical switchgear apparatus **10** shown in the previous figures being labeled with like reference numerals plus "1000."

[0081] FIG. **18** illustrates the electrical circuit diagram for the electrical switchgear apparatus **1010**. With reference to FIGS. **19-20**, the electrical switchgear apparatus **1010** includes an electrical switch **1200** coupled to the lid **1014** and electrically coupled to line couplers **1042** and input couplers **1046** into which fuses are inserted. The electrical switch **1200** includes an operation assembly **1208** positioned between the base plate **1224** and the lid **1014**. As shown in FIGS. **19** and **20**, the electrical switchgear apparatus **1010** is a one-phase switchgear apparatus that uses two switches (e.g., the stationary contact **264** and movable contact **268**), with a fuse coupled between the stationary contacts **264**.

[0082] FIGS. **21-23** illustrate an embodiment of an electrical switchgear apparatus **2010** (e.g., a 2-phase, fused electrical switchgear apparatus), with like components and features as the embodiments of the electrical switchgear apparatus **10** shown in the previous figures being labeled with like reference numerals plus "2000."

[0083] FIG. **21** illustrates the electrical circuit diagram for the electrical switchgear apparatus **2010**. With reference to FIGS. **22-23**, the electrical switchgear apparatus **2010** includes two electrical switches **2200a**, **2200b** coupled to the lid **2014** and electrically coupled to the line couplers **2042** and the input couplers **2046** into which fuses are inserted. The first electrical switch **2200a** includes an operation assembly **2208** positioned between the base plate **2224** and the lid **2014** and the chambers **2248** are positioned toward the center of the housing **2022**. The second electrical switch **2200b** includes an operation assembly **2350** with a transmission assembly **2352** positioned between the base plate **2224** and the lid **2014** and the chambers **2248** are also positioned toward the center of the housing **2022**.

[0084] FIGS. **24-26** illustrate an embodiment of an electrical switchgear apparatus **3010** (e.g., a 3-phase, fused electrical switchgear apparatus), with like components and features as the embodiments of the electrical switchgear apparatus **10** shown in the previous figures being labeled with like reference numerals plus "3000."

[0085] FIG. **24** illustrates the electrical circuit diagram for the electrical switchgear apparatus **3010**. With reference to FIGS. **25-26**, the electrical switchgear apparatus **3010** includes two electrical switches **3200a**, **3200b** coupled to the lid **3014** and electrically coupled to the line couplers **3042** and the input couplers **3046**, into which fuses are inserted. The first electrical switch **3200a** includes an operation assembly **3208** positioned between the base plate **3224** and the lid **3014** and the chambers **3248** are positioned toward the center of the housing **3022**. The second electrical

switch **3200b** includes an operation assembly **3350** with a transmission assembly **3352** positioned between the base plate **3224** and the lid **3014** and the chambers **3248** are also positioned toward the center of the housing **3022**.

[0086] FIGS. **27-29** illustrate an embodiment of an electrical switchgear apparatus **4010** (e.g., a 2-phase, non-fused, tapped electrical switchgear apparatus), with like components and features as the embodiments of the electrical switchgear apparatus **10** shown in the previous figures being labeled with like reference numerals plus “4000.”

[0087] FIG. **27** illustrates the electrical circuit diagram for the electrical switchgear apparatus **4010**. With reference to FIGS. **28-30**, the electrical switchgear apparatus **4010** includes two electrical switches **4200a**, **4200b** coupled to the lid **4014** and electrically coupled to the line couplers **4042** and the input couplers **4046** (e.g., voltage taps). The first electrical switch **4200a** includes an operation assembly **4208** positioned between the base plate **4224** and the lid **4014** and the chambers **4248** are positioned toward the center of the housing **4022**. The second electrical switch **4200b** includes an operation assembly **4350** with a transmission assembly **4352** positioned between the base plate **4224** and the lid **4014** and the chambers **4248** are also positioned toward the center of the housing **4022**.

[0088] FIGS. **30-32** illustrate an embodiment of an electrical switchgear apparatus **5010** (e.g., a 3-phase, non-fused, tapped electrical switchgear apparatus), with like components and features as the embodiments of the electrical switchgear apparatus **10** shown in the previous figures being labeled with like reference numerals plus “5000.”

[0089] FIG. **30** illustrates the electrical circuit diagram for the electrical switchgear apparatus **5010**. With reference to FIGS. **31-32**, the electrical switchgear apparatus **5010** includes two electrical switches **5200a**, **5200b** coupled to the lid **5014** and electrically coupled to the line couplers **5042** and the input couplers **5046** (e.g., voltage taps). The first electrical switch **5200a** includes an operation assembly **5208** positioned between the base plate **5224** and the lid **5014** and the chambers **5248** are positioned toward the center of the housing **5022**. The second electrical switch **4200b** includes an operation assembly **5350** with a transmission assembly **5352** positioned between the base plate **5224** and the lid **5014** and the chambers **5248** are also positioned toward the center of the housing **5022**.

[0090] FIGS. **33-35** illustrate an embodiment of an electrical switchgear apparatus **6010** (e.g., a 2-phase, non-fused electrical switchgear apparatus), with like components and features as the embodiments of the electrical switchgear apparatus **10** shown in the previous figures being labeled with like reference numerals plus “6000.”

[0091] FIG. **33** illustrates the electrical circuit diagram for the electrical switchgear apparatus **6010**. With reference to FIGS. **35-36**, the electrical switchgear apparatus **6010** includes an electrical switch **6200** coupled to the lid **6014** and electrically coupled to line couplers **6042**. The electrical switch **6200** includes an operation assembly **6350** with a transmission assembly **6352** positioned between the base plate **6224** and the lid **6014**.

[0092] FIGS. **36-38** illustrate an embodiment of an electrical switchgear apparatus **7010** (e.g., a 3-phase, non-fused electrical switchgear apparatus), with like components and features as the embodiments of the electrical switchgear apparatus **10** shown in the previous figures being labeled with like reference numerals plus “7000.”

[0093] FIG. **36** illustrates the electrical circuit diagram for the electrical switchgear apparatus **7010**. With reference to FIGS. **37-38**, the electrical switchgear apparatus **7010** includes an electrical switch **7200** coupled to the lid **7014** and electrically coupled to line couplers **7042**. The electrical switch **7200** includes an operation assembly **7350** with a transmission assembly **7352** positioned between the base plate **7224** and the lid **7014**.

[0094] Although the disclosure has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more

independent aspects of the disclosure as described.
[0095] Various features of the disclosure are set forth in the following claims.

Claims

1. A switchgear apparatus comprising: a cylindrical housing defining a housing interior filled with a non-SF6 gas; an electrical switch supported within the housing, the electrical switch including a chamber communicating the housing interior and a pair of contacts within the chamber, the pair of contacts including a stationary contact and a movable contact; a crankshaft coupled to the movable contact such that the movable contact is configured to translate linearly relative to the stationary contact in response to rotation of the crankshaft.
2. The switchgear apparatus of claim 1, wherein the electrical switch is a first electrical switch, and wherein the switchgear apparatus further comprises a second electrical switch supported within the housing.
3. The switchgear apparatus of claim 2, wherein the crankshaft is a first crankshaft, and wherein the switchgear apparatus further comprises a second crankshaft coupled to a movable contact of the second electrical switch.
4. The switchgear apparatus of claim 3, wherein the first crankshaft is rotatable in a first direction to move the movable contact of the first electrical switch away from the stationary contact of the first electrical switch, and wherein the second crankshaft is rotatable in a second direction to move the movable contact of the second electrical switch away from a stationary contact of the second electrical switch.
5. The switchgear apparatus of claim 4, wherein the second direction is opposite the first direction.
6. The switchgear apparatus of claim 1, wherein the non-SF6 gas is carbon dioxide.
7. The switchgear apparatus of claim 6, wherein the housing is filled with carbon dioxide to a gauge pressure of at least 15 psi.
8. The switchgear apparatus of claim 7, wherein the housing is filled with carbon dioxide to a gauge pressure of at least 20 psi.
9. The switchgear apparatus of claim 1, wherein the housing has an interior volume that is less than 500 liters.
10. The switchgear apparatus of claim 9, wherein the interior volume is less than 300 liters.
11. The switchgear apparatus of claim 10, wherein the interior volume is less than 150 liters.
12. A switchgear apparatus comprising: an apparatus housing including a base casting coupled to a shell, a top plate, and a base plate, the top plate and base plate coupled to opposite ends of the base casting; a crankshaft rotationally supported between the top plate and the base plate; an operation assembly configured to provide a rotational output to the crankshaft; a plurality of chambers coupled to the base casting and extending from the base casting away from the shell; a top support coupled to distal ends of the chambers, the top support configured to seal the distal ends of the chambers; a plurality of contact pairs supported in each chamber, each contact pair including a stationary contact and a movable contact, each stationary contact is coupled to the top support and each movable contact is coupled to the crankshaft, each movable contact configured to translate linearly along the chamber in response to rotation of the crankshaft.
13. The switchgear apparatus of claim 12, wherein each of the chambers is filled with a non-SF6 gas.
14. The switchgear apparatus of claim 12, wherein the operation assembly includes an input shaft coupled to the crankshaft and configured to provide the rotational output to the crankshaft, the input shaft configured to receive a rotational input from an operator, the rotational input having a first direction.
15. The switchgear apparatus of claim 14, wherein the operation assembly further includes a transmission assembly configured to receive the rotational input in the first direction and provide

the rotational output to the crankshaft in a second direction opposite to the first direction.

16. The switchgear apparatus of claim 15, wherein the transmission assembly includes a pinion coupled to the input shaft and a driven gear coupled to an output shaft, the output shaft coupled to the crankshaft.

17. A switchgear apparatus comprising: a housing defining a housing interior containing an insulating medium; a lid coupled to the housing and the lid defining an interior; a plurality of connectors extending from the lid into the interior; and an electrical switch coupled to the lid and electrically coupled to the connectors, the electrical switch including an apparatus housing including a base casting coupled to a shell, a top plate, and a base plate, the top plate and base plate coupled to opposite ends of the base casting, a crankshaft rotationally supported between the top plate and the base plate, an operation assembly coupled to the base plate and configured to provide a rotational output to the crankshaft, a plurality of chambers coupled to the base casting and extending from the base casting away from the shell, a top support coupled to distal ends of the chambers, the top support configured to seal the distal ends of the chambers, and a plurality of contact pairs supported in each chamber, each contact pair including a stationary contact and a movable contact, each stationary contact is coupled to the top support and each movable contact is coupled to the crankshaft, each movable contact configured to translate linearly along the chamber in response to rotation of the crankshaft.

18. The switchgear apparatus of claim 17, wherein the electrical switch is a first electrical switch, the switchgear apparatus including a second electrical switch.

19. The switchgear apparatus of claim 17, wherein the operation assembly includes a transmission assembly configured to receive a rotational input in a first direction and provide the rotational output to the crankshaft in a second direction opposite to the first direction.

20. The switchgear apparatus of claim 19, wherein the transmission assembly includes a pinion coupled to an input shaft and a driven gear coupled to an output shaft, the output shaft coupled to the crankshaft.
